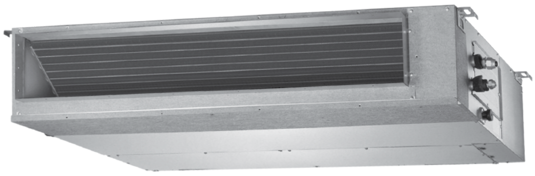
LOW PROFILE SPLIT SYSTEM

10.00 kW | 1 Phase | 1 Stage



UNIT FEATURES

- Reverse Cycle Low Profile Split System
- Twin-Rotary Compressor
- Superior Operating Range:
  - Cooling: up to 50°C DB
  - Heating: down to -25°C DB
- Adjustable Airflow
- Fan Speed: Auto, Low, Medium and High
- External Static Pressure Settings
- Powder Coated Panels - Outdoor Unit
- Hydrophilic Indoor and Outdoor Coil Protection
- Self-Diagnosis and Auto Protection
- Fire Proof Electrical Box - Indoor & Outdoor
- Return Air Sensor Included in the Unit
- Drain Pump
- Hanging Brackets

UNIT OPTION

- Left or Right Hand Drain Connection
- Bottom Return Air Connection
- Outside Air Intake

CONTROL FEATURES

- 7-Day Scheduler/Weekly Timer
- Digital Display
- Wired Controller
- Remote Controller - Optional
- Auto Restart After Power Failure
- Timer Operation
- Remote ON/OFF Input
- 4-Speed Indoor Fan
- 5-Speed Outdoor Fan
- Sleep Mode
- Turbo Mode
- Dry Mode Operation
- Demand Response Ready
- Auto Defrost Function
- Follow Me Function
- Fault Alarm Output

UNIT COMPLIANCE

- MEPS 2012
- Demand Response AS4755.3.1

SPECIFICATION SUMMARY

OUTDOOR UNIT MODEL		URC-100AS
INDOOR UNIT MODEL		LRE-100AS
		NETT
[1][2] COOLING CAPACITY (kW) - NOMINAL (MIN - MAX)		10.00 (4.00 - 12.00)
[1][3] HEATING CAPACITY (kW) - NOMINAL (MIN - MAX)		10.90 (4.20 - 12.50)
[1][4] COOLING INPUT POWER (kW)		3.02
[1][4] HEATING INPUT POWER (kW)		3.05
[1][2] EER		3.31
[1][3] COP		3.57
[5] INDOOR AIRFLOW (l/s) - MIN/NOMINAL/MAX		440/520/600
MOISTURE REMOVAL (l/hr)		1.7
INDOOR SOUND PRESS. LEVEL dB(A) - MIN/NOMINAL/MAX @ SP1		40/46/48
OUTDOOR SOUND PRESS. LEVEL @ 1M dB(A)		62
[6] OUTDOOR SOUND POWER LEVEL dB(A)		70
POWER SUPPLY		220-240V/1Ph+N/50 Hz
INDOOR UNIT WIRING METHOD		Hard wire to Outdoor
[1] RATED LOAD AMPS - COOLING / HEATING		14.1 / 13.7
[7] FULL LOAD AMPS		21.5
[8] CIRCUIT BREAKER AND CABLE AMPS		25.0
WEIGHT (kg) - INDOOR / OUTDOOR		40 / 69
OUTDOOR OPERATING RANGE (ºC)	COOLING	-15 to 50
	HEATING	-25 to 30

[1] Measured and tested in accordance with AS/NZS 3823.1.2.
[2] At 27ºC DB / 19ºC WB entering air temperatures and 35ºC ambient.
[3] At 20ºC DB entering air temperature and 7ºC DB / 6ºC WB ambient.
[4] input power includes indoor fan kW.
[5] Max. - Min. airflow application range.
[6] Determination of Sound Power Levels of Noise Sources per AS1217.2.
[7] Full Load Amps are based on compressor and fan motors' maximum expected current.
[8] See Specifications sheet for cable size and circuit breaker size details.

Note: Use input power to estimate running cost.

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CAPACITY SELECTION DATA | URC-100AS / LRE-100AS

10.00 kW | 1 Phase | 1 Stage

COOLING PERFORMANCE

INDOOR CONDITIONS		OUTDOOR TEMPERATURE (°C - DB)						
		21	25	30	35	40	45	50
21°C - DB 15°C - WB	Nett Capacity, kW	9.73	9.56	9.39	8.93	8.53	6.25	4.47
	Sensible Heat Capacity, kW	7.66	7.52	7.39	7.03	6.71	4.92	3.51
	Power Input, kW	1.16	1.91	2.32	2.84	3.18	3.21	3.24
24°C - DB 17°C - WB	Nett Capacity, kW	10.36	10.17	9.99	9.50	9.07	6.65	4.75
	Sensible Heat Capacity, kW	8.15	8.00	7.87	7.48	7.14	5.23	3.74
	Power Input, kW	1.19	1.97	2.40	2.93	3.28	3.31	3.34
27°C - DB 19°C - WB	Nett Capacity, kW	10.90	10.70	10.52	**10.00**	9.55	7.00	5.00
	Sensible Heat Capacity, kW	8.58	8.42	8.28	**7.87**	7.52	5.51	3.94
	Power Input, kW	1.23	2.03	2.47	**3.02**	3.38	3.41	3.45
32°C - DB 23°C - WB	Nett Capacity, kW	12.21	11.98	11.78	11.20	10.70	7.84	5.60
	Sensible Heat Capacity, kW	9.61	9.43	9.27	8.81	8.42	6.17	4.41
	Power Input, kW	1.36	2.24	2.73	3.33	3.73	3.77	3.81

HEATING PERFORMANCE

INDOOR CONDITIONS		OUTDOOR TEMPERATURE							
		-15°C D -16°C W	-7°C D -8°C W	-5°C D -6°C W	0°C D -1°C W	4°C D 3°C W	7°C D 6°C W	12°C D 11°C W	24°C D 18°C W
15°C - DB	Nett Capacity, kW	4.27	7.75	8.55	9.46	10.07	12.21	13.43	11.26
	Power Input, kW	1.72	2.79	2.46	3.01	3.38	3.35	3.68	3.08
18°C - DB	Nett Capacity, kW	4.08	7.41	8.16	9.04	9.62	11.66	12.83	10.76
	Power Input, kW	1.66	2.69	2.37	2.91	3.26	3.23	3.55	2.97
20°C - DB	Nett Capacity, kW	3.82	6.92	7.63	8.45	8.99	**10.90**	11.99	10.06
	Power Input, kW	1.56	2.54	2.24	2.75	3.08	**3.05**	3.36	2.81
22°C - DB	Nett Capacity, kW	3.70	6.71	7.40	8.19	8.72	10.57	11.63	9.75
	Power Input, kW	1.60	2.59	2.28	2.80	3.15	3.11	3.43	2.87
27°C - DB	Nett Capacity, kW	3.32	6.02	6.64	7.35	7.82	9.48	10.43	8.75
	Power Input, kW	1.60	2.60	2.29	2.81	3.15	3.12	3.43	2.87

PIPE LENGTH CORRECTION MULTIPLIER

COOLING			PIPE LENGTH (m)							
			5	10	20	30	40	50	60	65
H = Height Difference (m)	Indoor Unit Higher Than Outdoor Unit	30	---	---	---	0.891	0.867	0.843	0.818	0.806
		20	---	---	0.926	0.904	0.880	0.855	0.831	0.819
		10	---	0.970	0.940	0.918	0.893	0.868	0.844	0.831
		5	0.995	0.980	0.950	0.927	0.902	0.877	0.852	0.840
		0	1.000	0.985	0.954	0.932	0.907	0.882	0.856	0.844
	Indoor Unit Lower Than Outdoor Unit	-5	0.999	0.984	0.953	0.931	0.906	0.881	0.856	0.843
		-10	---	0.983	0.952	0.930	0.905	0.880	0.855	0.842
		-20	---	---	0.952	0.929	0.904	0.879	0.854	0.841
		-30	---	---	---	0.928	0.903	0.878	0.853	0.840

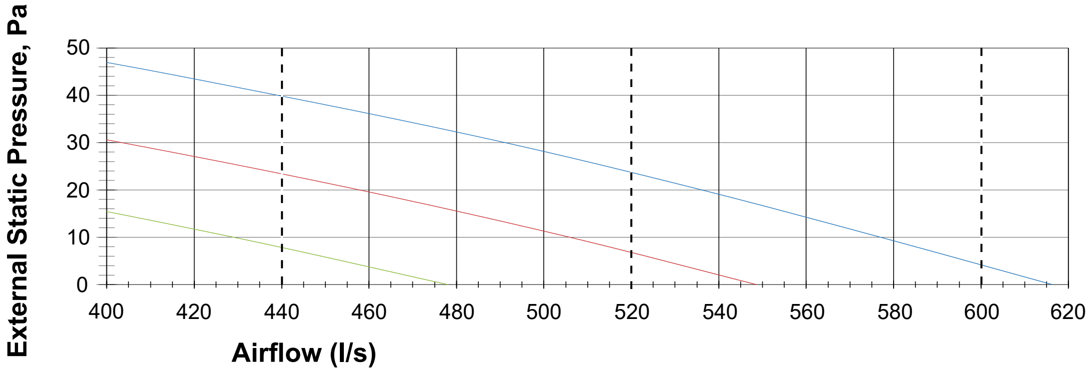
HEATING			PIPE LENGTH (m)							
			5	10	20	30	40	50	60	65
H = Height Difference (m)	Indoor Unit Higher Than Outdoor Unit	30	---	---	---	0.965	0.952	0.938	0.924	0.918
		20	---	---	0.968	0.966	0.953	0.939	0.925	0.919
		10	---	0.988	0.969	0.967	0.953	0.940	0.926	0.920
		5	0.999	0.989	0.970	0.968	0.954	0.941	0.927	0.920
		0	**1.000**	**0.990**	**0.971**	**0.969**	**0.955**	**0.942**	**0.928**	**0.921**
	Indoor Unit Lower Than Outdoor Unit	-5	0.998	0.988	0.969	0.967	0.953	0.940	0.926	0.920
		-10	---	0.986	0.967	0.965	0.952	0.938	0.924	0.918
		-20	---	---	0.965	0.963	0.950	0.936	0.923	0.916
		-30	---	---	---	0.961	0.948	0.934	0.921	0.914

10.00 kW

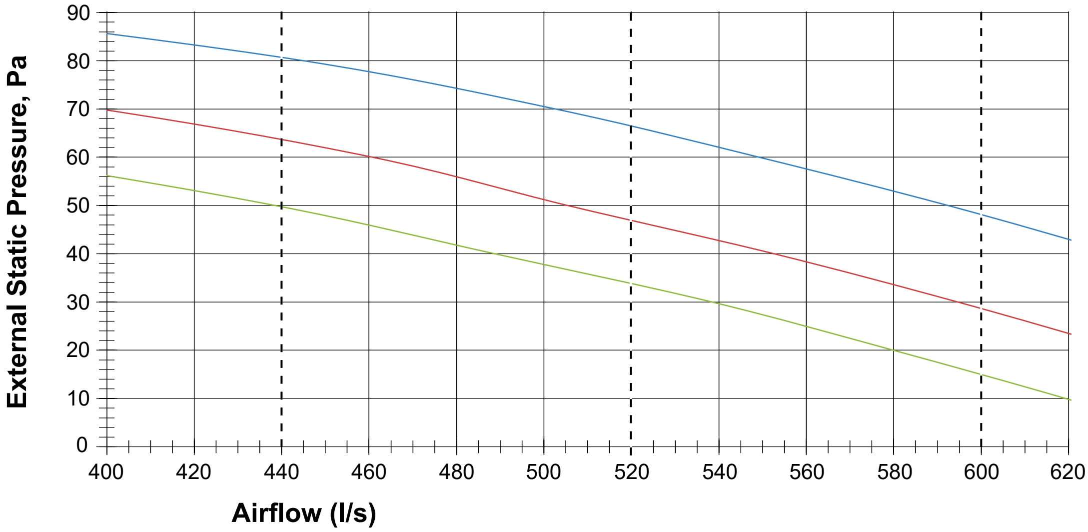
1 Phase | 1 Stage

*Default Settings

SP1*



SP2



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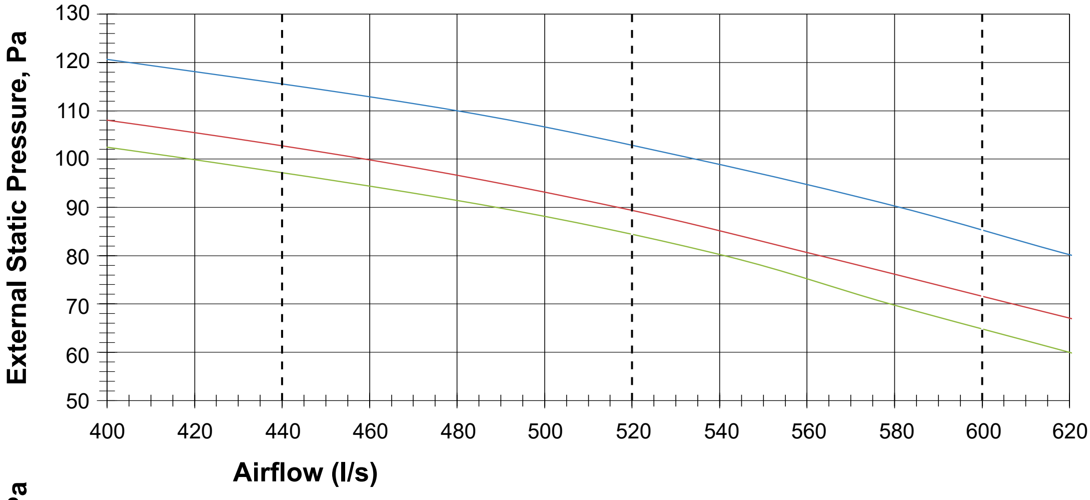

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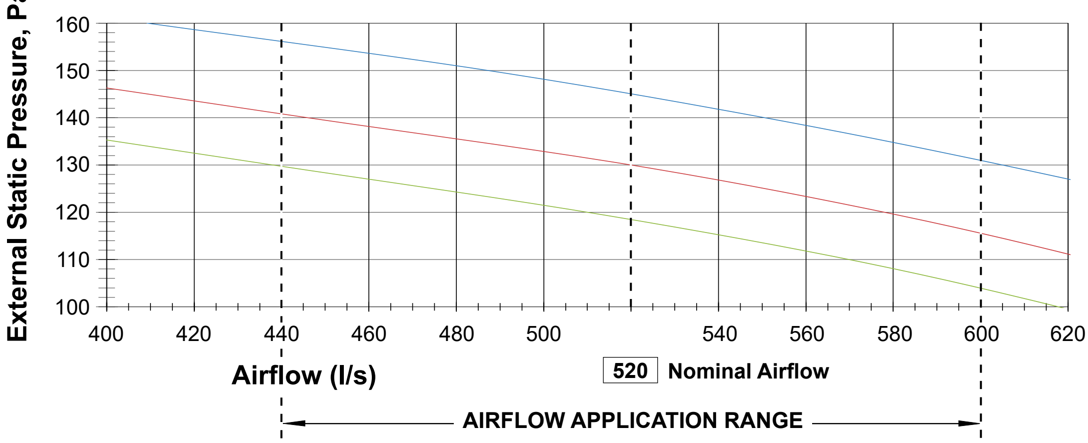
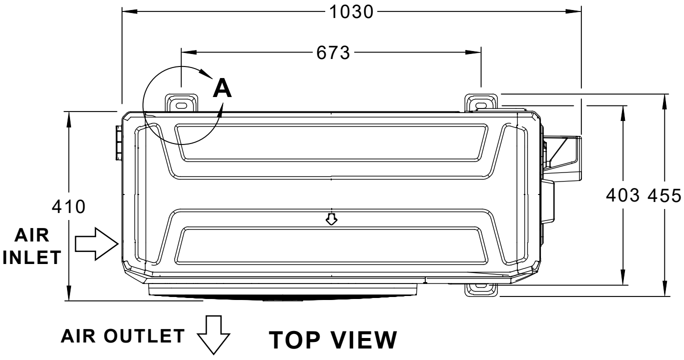


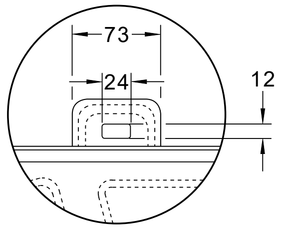
UNIT DIMENSIONS — URC-100AS / LRE-100AS

C OUTDOOR UNIT: URC-100AS

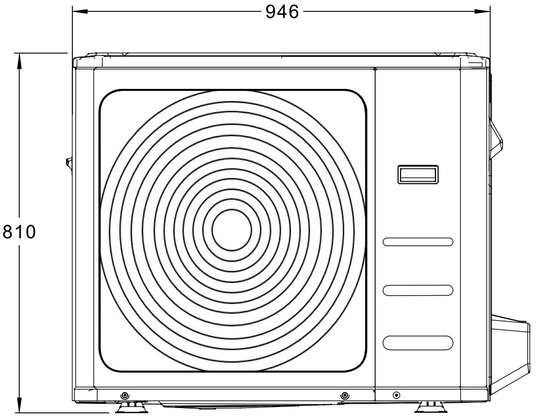
NOMINAL DIMENSION (H x W x D)
= 810 x 946 x 410
USE M10 BOLT FOR FEET MOUNTING



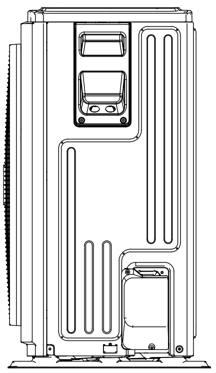
TOP VIEW



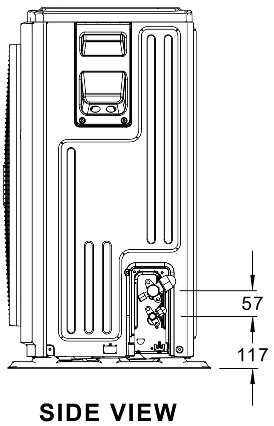
DETAIL - "A"



FRONT VIEW



SIDE VIEW



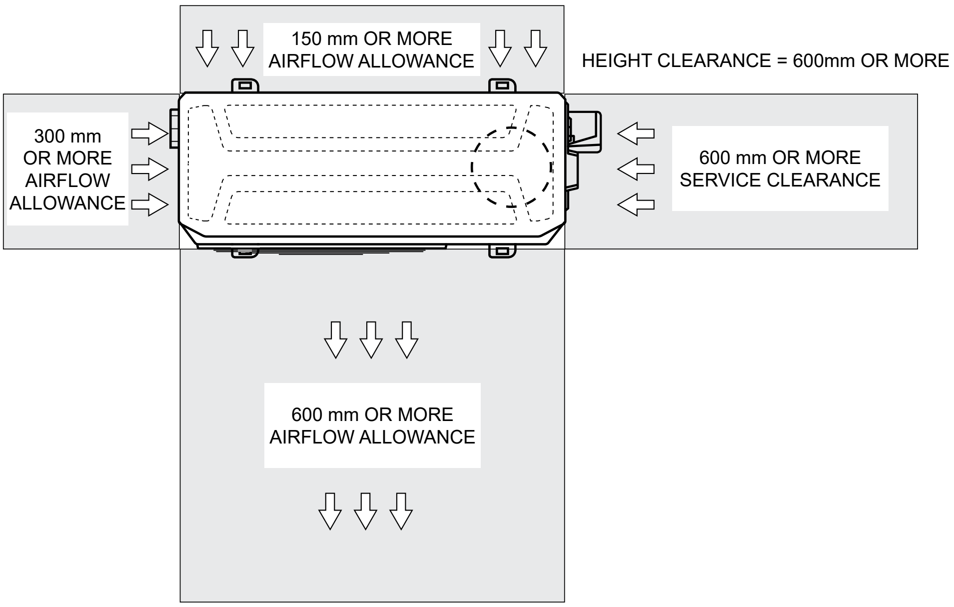
SIDE VIEW

NOTES:
1. All dimensions are in mm unless specified.
2. Do not scale drawing.
3. Refer Pipe Connection Details on Specifications Sheet.
4. Suggested Service Clearances are based on conditions that the spaces are free from obstructions and walkway passage of 1000mm is available.
5. Minimum service access areas are responsibilities of the installer.

THIRD ANGLE PROJECTION

PLEASE NOTE THAT UNDER ALL CIRCUMSTANCES, CONDENSER AIR MUST NOT RECIRCULATE BACK ONTO CONDENSER COIL.

SERVICE ACCESS AREAS & AIRFLOW ALLOWANCES



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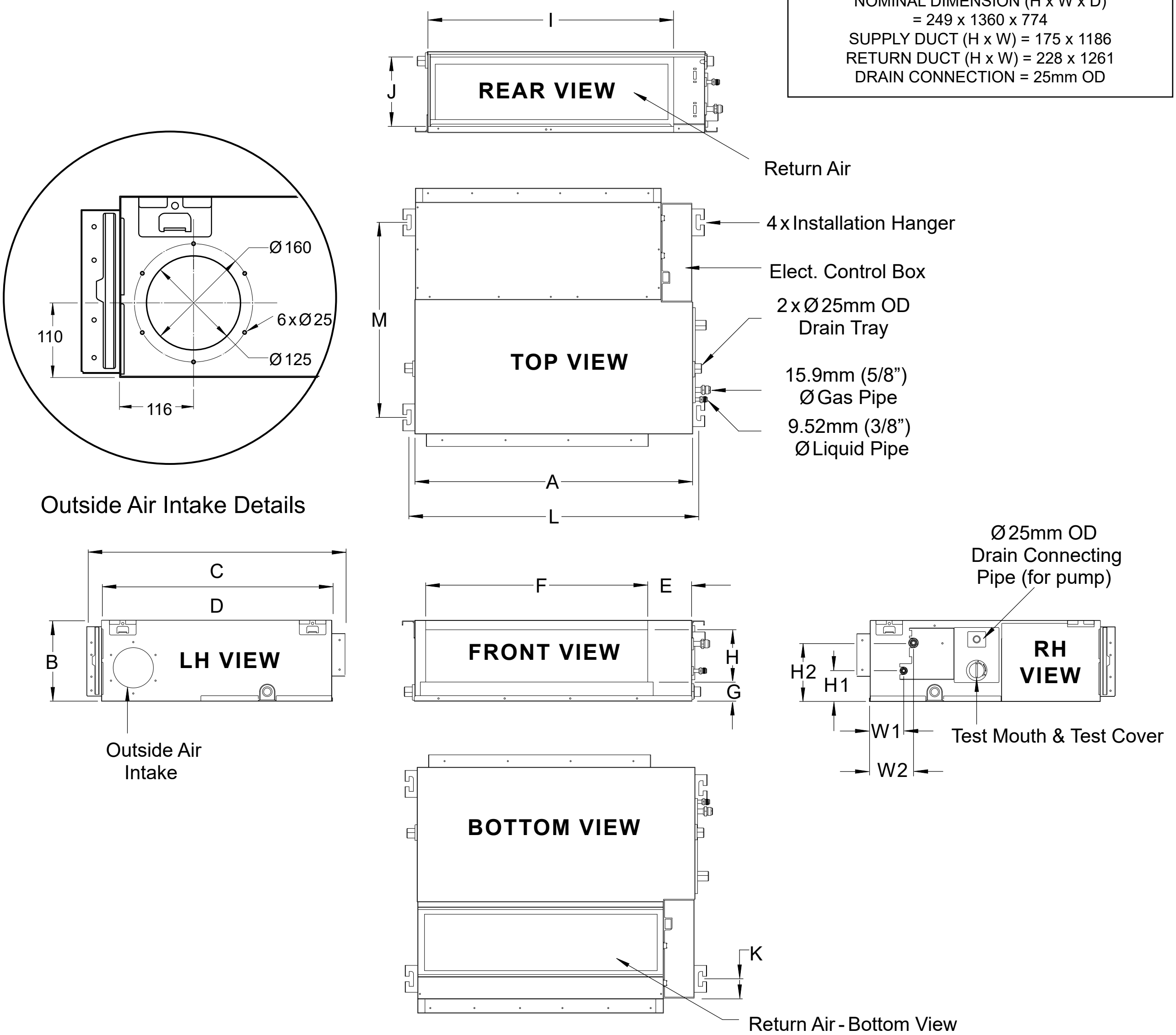
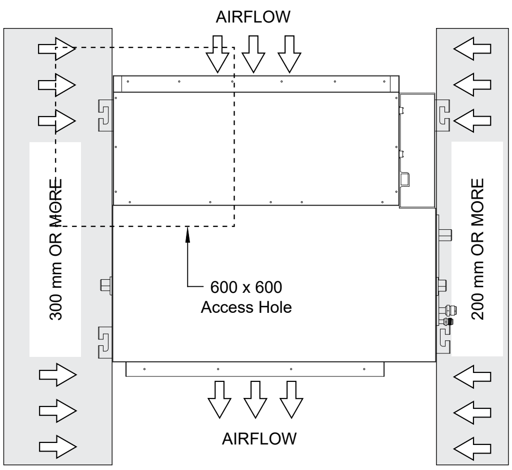
E INDOOR UNIT: LRE-100AS

NOMINAL DIMENSION (H x W x D)
= 249 x 1360 x 774
SUPPLY DUCT (H x W) = 175 x 1186
RETURN DUCT (H x W) = 228 x 1261
DRAIN CONNECTION = 25mm OD



Dimensions (mm)																
A	B	C	D	E	F	G	H	I	J	K	L	M	H1	H2	W1	W2
1360	249	774	700	140	1186	50	175	1261	228	50	1400	598	80	150	130	155

SERVICE ACCESS AREAS & AIRFLOW ALLOWANCES



NOTES:

THIRD ANGLE PROJECTION

1. All dimensions are in mm unless specified.
2. Do not scale drawing.
3. Refer Pipe Connection Details on Specifications Sheet.
4. Minimum service access areas are responsibilities of the installer.
5. The service areas & airflow allowances mentioned are the minimum requirements to complete minor repairs. Additional space may be required should a major repair need to be undertaken.

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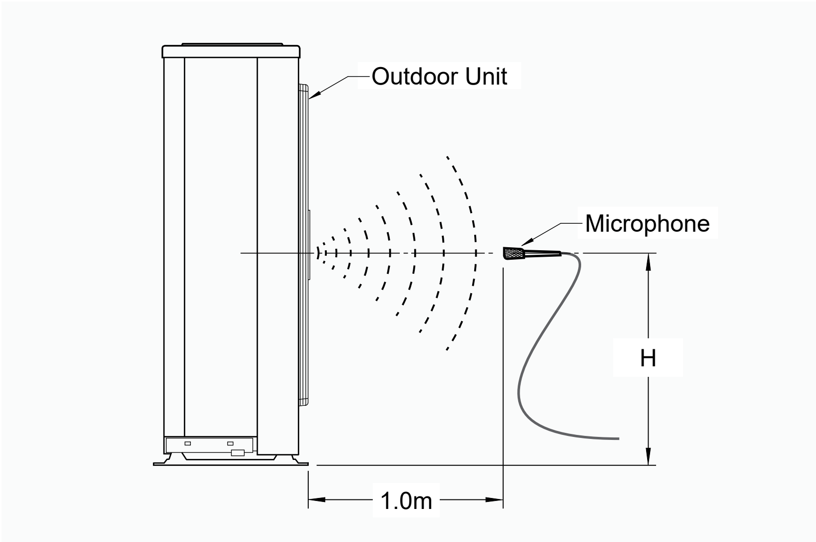
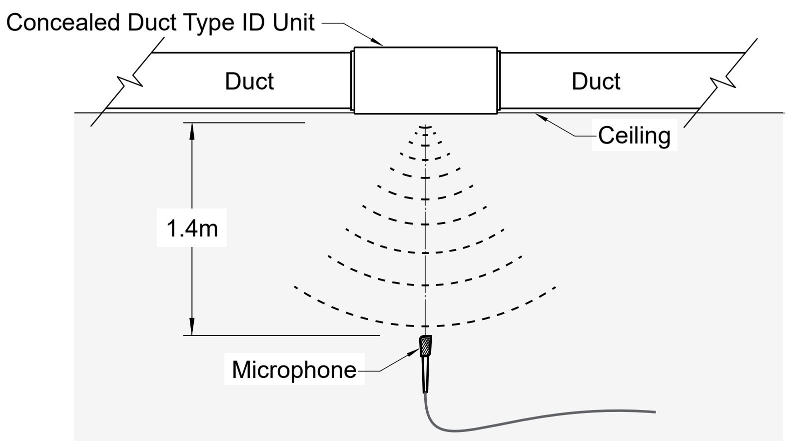
OUTDOOR UNIT: URC-100AS



Model	Sound Pressure Level dB(A)
URC-100AS	62

NOTE: H= 0.5 x Height of Outdoor Unit.

INDOOR UNIT: LRE-100AS



Model	Sound Pressure Level dB(A)		
	High	Medium	Low
LRE-100AS	48	46	40

SPECIFICATIONS

URC-100AS / LRE-100AS

1 Phase | 1 Stage | 10.00 kW

UNIT DIMENSIONS

OUTDOOR DIMENSIONS	Depth	410 mm
	Height	810 mm
	Width	946 mm
INDOOR DIMENSIONS	Depth	774 mm
	Height	249 mm
	Width	1360 mm

ELECTRICAL

POWER SUPPLY		220 - 240 Volts / 1 Ph + N / 50Hz
WIRING METHOD		Hard wire to outdoor
FULL LOAD AMPS*	Total	21.5
FULL LOAD AMPS	Indoor	2.8
RATED LOAD AMPS**	Cooling	14.1
	Heating	13.7
IP RATING	Outdoor	IP24
	Indoor	IP20

IMPORTANT - The local electricity authority may require limits on starting current and voltage drop, please check prior to purchase.

*Full Load Amps are based on Compressor and Fan Motor's maximum expected current.

**Rated Load Amps are measured and tested in accordance with AS/NZS3823.1.1.

CABLE SIZE & CIRCUIT BREAKER SIZE

Suggested minimum cable size should be used as a guide only, refer to AS/NZS 3000 "Australian/New Zealand Wiring Rules" for more details.

Cable Size (Supply Mains)	4.0 mm^2 (SUGGESTED MINIMUM)
Cable Size (Indoor to Outdoor Wire)	1.0 mm^2 (2 Core + Earth)
Circuit Breaker Size	25.0 Amps

OUTDOOR COIL

TUBE TYPE	Copper Ø7mm, inner groove tube
FIN TYPE	Hydrophylic Aluminium
FACE AREA	0.76 m^2
FIN SPACING	1.4 mm
ROWS	3

OUTDOOR FAN

NUMBER OF FANS x TYPE	1 x Axial
INPUT (W)	150
AIRFLOW (l/s)	1069

INDOOR COIL

TUBE TYPE	Copper Ø7mm, inner groove tube
FIN TYPE	Hydrophylic Aluminium
FACE AREA	0.345 m^2
FIN SPACING	1.4 mm
ROWS	4

INDOOR FAN

NUMBER OF FANS x TYPE	1 x Centrifugal
INPUT (W)	420
AIRFLOW (l/s) - Min / Nom / Max	440 / 520 / 600

AIR FILTERS

All return air including fresh air must have adequate filter supplied and fitted by installating contractor. Filters must be always accessible for cleaning. ActronAir does not supply or make any provisions for return air filter.

COMPRESSOR

NUMBER PER UNIT x TYPE	1 x Twin-Rotary Compressor
STARTING METHOD	DC Inverter Starter
INPUT (W)	2600
REFRIGERANT OIL (TYPE / CHARGE)	ESTER OIL VG74 / 1000ml
PROTECTION	External Thermal Cut-Out

REFRIGERATION SYSTEM

REFRIGERANT TYPE	R-410A
FACTORY CHARGE	3800 g
PRE-CHARGE LENGTH	15 m
ADD'L. REFRIGERANT CHARGE	30 g / m
DESIGN PRESSURE (High / Low)	4.2 / 1.5 MPa

INTERCONNECTING PIPE RUN

MAX PIPE LENGTH		65 m
MIN PIPE LENGTH		5 m
MAX. VERTICAL LENGTH		30 m (Included in Max. Pipe Length)
FIELD PIPE SIZES		
Liquid Pipe		9.52 mm (3/8")
Gas Pipe		15.9 mm (5/8")
PIPE CONNECTIONS		
Indoor	Liquid Pipe	9.52 mm (3/8")
	Gas Pipe	15.9 mm (5/8")
Outdoor	Liquid Pipe	9.52 mm (3/8")
	Gas Pipe	15.9 mm (5/8")
CONNECTION TYPE		Flare Nut

ELECTRIC CONTROLS

DEFROST METHOD	Reverse Cycle
CONTROL FIELD WIRING (OUTDOOR TO INDOOR-FIELD SUPPLIED)	2 Core 14 / 0.20 (0.44mm^2) Shielded Data Cable
WALL CONTROLLER CABLE (INCLUDED WITH WALL CONTROLLER)	4 Core (0.75mm^2) Shielded Data Cable

OPERATING RANGE

It is essential that the unit is correctly sized for the application and operates within its recommended range of operating conditions as shown below.

MODE	RANGE	INDOOR OPERATING TEMPERATURE	OUTDOOR AIR INTAKE TEMPERATURE
COOLING	Max.	32°C DB	50°C DB
	Min.	17°C DB	-15°C DB
HEATING	Max.	30°C DB	30°C DB
	Min.	0°C DB	-25°C DB

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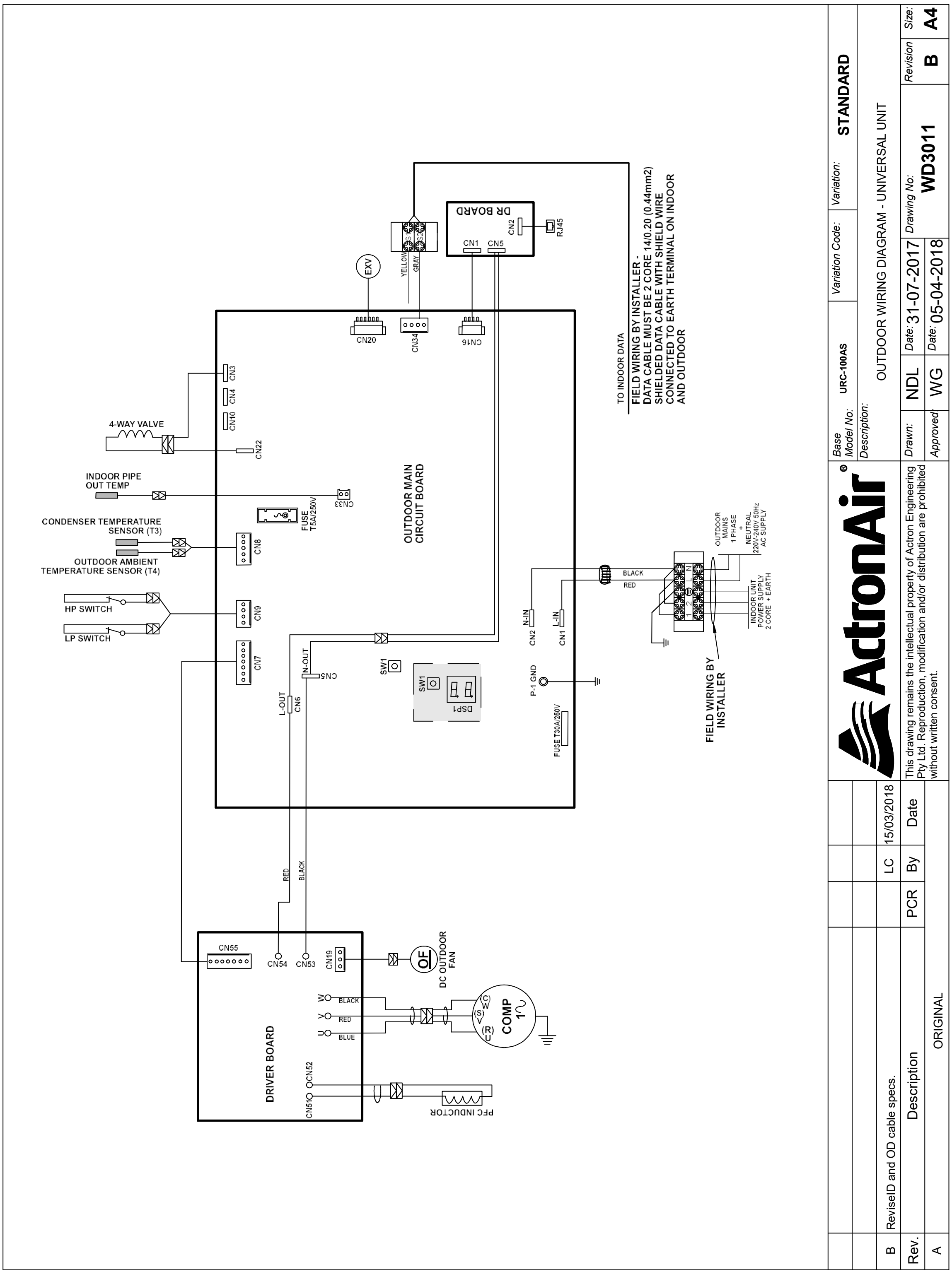
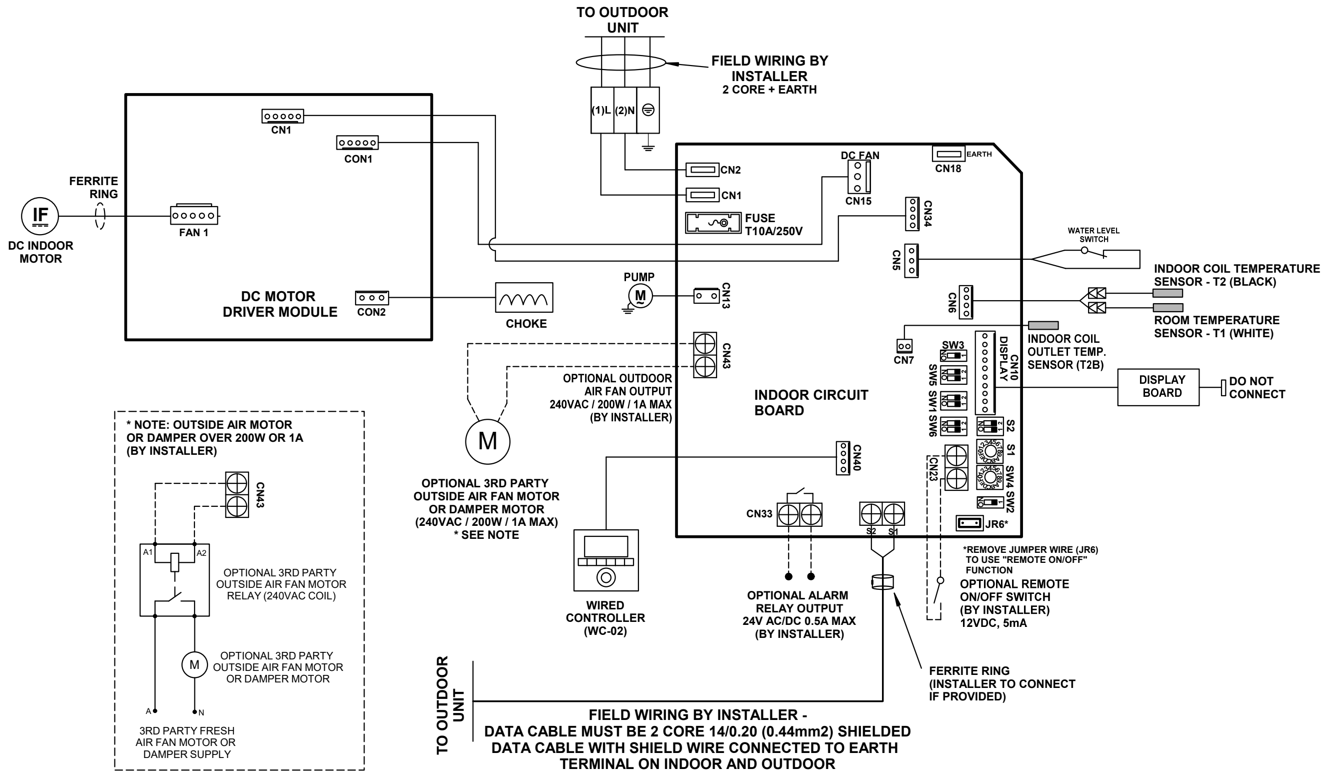
URC-100AS (OUTDOOR)



WIRING DIAGRAM

URC-100AS / LRE-100AS

LRE-100AS (INDOOR)



					ActronAir®	Base Model No: LRE-071AS/ LRE-100AS/ LRE-130AS		Variation Code:	Variation: STANDARD	
						Description: INDOOR WIRING DIAGRAM - SUPER SLIM				
B	ReviseID and OD cable specs.		LC	15/03/2018						
Rev.	Description	PCR	By	Date	This drawing remains the intellectual property of Actron Engineering Pty Ltd. Reproduction, modification and/or distribution are prohibited without written consent.	Drawn: NDL	Date: 18-07-2017	Drawing No: WD3000	Revision B	Size: A4
A	ORIGINAL					Approved: WG	Date: 04-05-2018			

ActronAir

Ver. 7 190329

10.00 KW

1 Phase | 1 Stage